Prosociality by Coupling, Not Mere Observation: Homeostatic Sharing in an Inspectable Recurrent Artificial Life Agent

Aishik Sanyal[1]

[1]Independent Research Engineer
aishik@xcellect.com

Abstract

Artificial agents can be made to "help" for many reasons, including explicit social reward, hard-coded prosocial bonuses, or direct access to another agent's internal state. Those possibilities make minimal prosocial behavior hard to interpret. Building on ReCoN-Ipsundrum, an inspectable recurrent controller with affect-coupled regulation, I add an explicit homeostat and a social coupling channel while keeping planning strictly self-directed: the agent scores only its own predicted internal state, and no partner-welfare reward term is introduced. The study should therefore be read as a minimal mechanistic replication-and-decomposition of the observation-versus-coupling distinction in a hand-specified recurrent controller. I then compare four matched conditions in two toy worlds. In a one-step *FoodShareToy*, an exact solver finds a sharp switch from EAT to PASS at $\lambda^* \approx 0.91$ for the default state. In the experimental runs, `social_none` and `social_cognitive_direct` never help, whereas `social_affective_direct` and `social_full_direct` always do. In a multi-step *SocialCorridorWorld*, the same dissociation reappears: coupling flips help rate from 0 to 1, partner recovery from 0 to 1, and rescue latency from 18 to 9 steps, while raising mutual viability from 0.15 to 0.33. Sham lesions preserve helping, but coupling-off and shuffled-partner lesions abolish it in both tasks. A coupling sweep reveals a feasibility boundary shaped by coupling and metabolic load: under low load, helping appears for $\lambda \geq 0.25$, whereas under medium and high loads no tested coupling value rescues the partner within horizon. The result is a narrow but useful claim for artificial life: in this minimal architecture, mere partner-state access is not enough; helping appears when another's need is routed into self-regulation.

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Data/Code available at: github.com/xcellect/recips

Introduction

Artificial life is especially good at turning philosophical distinctions into executable ones. Braitenberg's vehicles showed how minimal mechanisms tempt us into psychological description (Braitenberg, 1984). Dennett described ALife as a form of "prosthetic imagination" for philosophy (Dennett, 1994). Pattee argued that such models matter only if we stay clear about what exactly their synthetic regularities are evidence for (Pattee, 1995). Those methodological concerns become particularly sharp when an agent appears to do something socially significant, such as helping another agent at a cost to itself.

In synthetic systems, "prosociality" is easy to trivialize. An agent may help because its reward function already contains another agent's welfare, because its action space directly privileges a helping move, or because the task is so underspecified that social interpretation outruns mechanism. Recent work by Yoshida and Man (2025) sharpened this problem by contrasting cognitive access to a partner's need with affective homeostatic coupling in learned multi-agent systems. Their central result is strong: prosocial behavior appears when the distress of another agent affects one's own regulation, not when that distress is merely observed.

Here I ask whether the same dissociation can be recovered in a much smaller and more inspectable setting. The aim is not to claim a new empirical discovery about prosociality, but to recover and decompose that distinction in a smaller executable object whose state updates, rollouts, and lesions can all be inspected directly. I extend Sanyal (2026), which introduced a recurrent ReCoN-based controller with an ipsundrum-style persistence loop and optional affect-coupled regulation. The extension keeps the same broad control logic but adds an explicit scalar homeostat and a social coupling pathway. The key design constraint is theoretical rather than performance-driven: the planner remains self-directed. It scores only the agent's own predicted internal state after rollout. No external reward bonus for partner welfare is added, and no direct "help partner" term is introduced.

That design lets me make a cleaner claim. If helping appears, it must do so because the partner's distress has been routed into the agent's own control problem. If helping disappears under lesion, that disappearance can be tied to a specific channel rather than to a change in task reward. This is a narrower claim than empathy, consciousness, or moral status. It is a claim about control architecture.

The contribution is therefore a mechanistic artificial life test rather than a benchmark. More specifically, this paper is

a minimal mechanistic replication-and-decomposition of the observation-versus-coupling distinction in a hand-specified recurrent ALife controller. I show: first, an exact one-step threshold in a food-sharing toy world; second, a matching multi-step rescue pattern in a corridor world; third, causal lesions that remove helping while leaving the rest of the controller intact; and fourth, a coupling sweep showing that prosociality depends on the relation between coupling strength and ecological load. The broader lesson is methodological. For ALife, tunability is not only a nuisance to control away. It is also the experimental handle that lets us ask which social descriptions survive mechanistic decomposition.

Mechanism and Experimental Design

Base controller and social extension

The starting point is ReCoN-Ipsundrum (Sanyal, 2026): a small ReCoN controller (Bach and Herger, 2015) augmented with a recurrent persistence loop inspired by Humphrey's ipsundrum hypothesis (Humphrey, 2023) and a minimal affect proxy inspired by constructionist affect (Barrett, 2017). The present extension keeps that recurrent-affective backbone but adds an explicit homeostatic resource state for each agent:

$$
\begin{aligned}
E_{t+1}^{\text{true}} &= \text{clip}\big(E_t^{\text{true}} - c_b - c_m a_t - c_h h_t + g_e e_t + g_p p_t, 0, 1\big),\\
d_t^{\text{self}} &= \max(0, s - E_t^{\text{model}}),\\
d_t^{\text{other}} &= \max(0, s - \hat{E}_t^{\text{other}}),\\
d_t^{\text{cpl}} &= d_t^{\text{self}} + \lambda\, d_t^{\text{other}},\\
E_t^{\text{pred}} &= \text{clip}\big(E_t^{\text{model}} - k_h d_t^{\text{cpl}}, 0, 1\big),\\
\text{PE}_t &= E_{t+1}^{\text{true}} - E_t^{\text{pred}},\\
E_{t+1}^{\text{model}} &= \text{clip}\big(E_t^{\text{model}} + k_{pe}\,\text{PE}_t, 0, 1\big).
\end{aligned}
\tag{1}
$$

Here s is the homeostatic setpoint, λ is affective coupling strength, and $\hat{E}^{\text{other}}$ is the partner-energy estimate. Valence and arousal are then computed from coupled distress and prediction error, and fed back into the existing recurrent loop just as in the parent architecture.

The crucial architectural constraint is that the policy remains self-directed. Candidate actions are evaluated by finite-horizon internal rollout, and the score contains only the agent's own predicted valence, arousal, recurrent salience N^s, and body-budget error. In the social tasks used in this work, those score weights are fixed at $w_{\text{valence}} = 2.0$, $w_{\text{arousal}} = -1.2$, $w_{N^s} = -0.8$, and $w_{\text{bb_err}} = -0.4$, with no separate progress, scenic-value, or novelty bonus. There is no separate social reward. Partner state can matter only by changing the actor's own coupled distress before scoring.

Table 1 defines the four direct-state conditions used in this work. In the current implementation, the same affect-capable base controller is reused throughout, and the cognitive condition should be read primarily as a control for partner-state access rather than as a richer social-inference model. Because $\lambda = 0$ removes the only route by which partner state can alter the self-directed score in this direct-state setup, the negative result below should be read as an architectural routing result rather than as a general theorem about social cognition. Scenic-value and novelty terms are disabled in the social tasks, and partner death does not terminate the actor's episode, preventing hidden incentives through task termination.

Condition	Partner access	Coupling
`none`	no	$\lambda = 0$
`cognitive`	direct	$\lambda = 0$
`affective`	direct to homeostat	$\lambda > 0$
`full`	direct + coupled	$\lambda > 0$

Table 1: Matched social conditions. The base recurrent-affective controller, action space, and planner are held fixed across conditions.

Tasks, lesions, and metrics

I use two deliberately small environments.

FoodShareToy. A possessor starts with one food item and chooses among EAT, PASS, and STAY. The partner is passive. Default initial energies are 0.55 for the possessor and 0.20 for the partner. Because the task is one-step, it admits an exact state solver using the same forward model and scorer as the policy.

SocialCorridorWorld. The possessor starts in the middle of a 1-D corridor. Food lies at the left end, the partner at the right end, and a hazard sits in the center corridor. Available actions are LEFT, RIGHT, GET, EAT, PASS, and STAY. Helping requires a full fetch–carry–pass sequence. The social planner evaluates action sequences with horizon 10 inside an 18-step episode.

Lesions. I test three lesion modes in the affective and full conditions: *sham*, *coupling_off*, and *shuffle_partner*. The last condition routes an incorrect partner-energy signal into the coupling channel, checking whether the policy depends on a veridical partner signal rather than on the mere presence of an extra scalar.

Coupling sweep. In the corridor task I sweep $\lambda \in \{0, 0.25, 0.5, 0.75, 1.0\}$ under low, medium, and high metabolic load. Load changes basal and movement costs as well as food gains, so the sweep tests ecological dependence rather than a single free parameter in isolation.

Readouts. The headline metrics are help rate when the partner is distressed, partner recovery rate, mutual viability, rescue latency, self-cost of help, and final energy. Each experimental condition is rerun over 64 nominal seeds as a reproducibility check. Under the validated setup, however, the trajectories are deterministic, so those reruns are not used

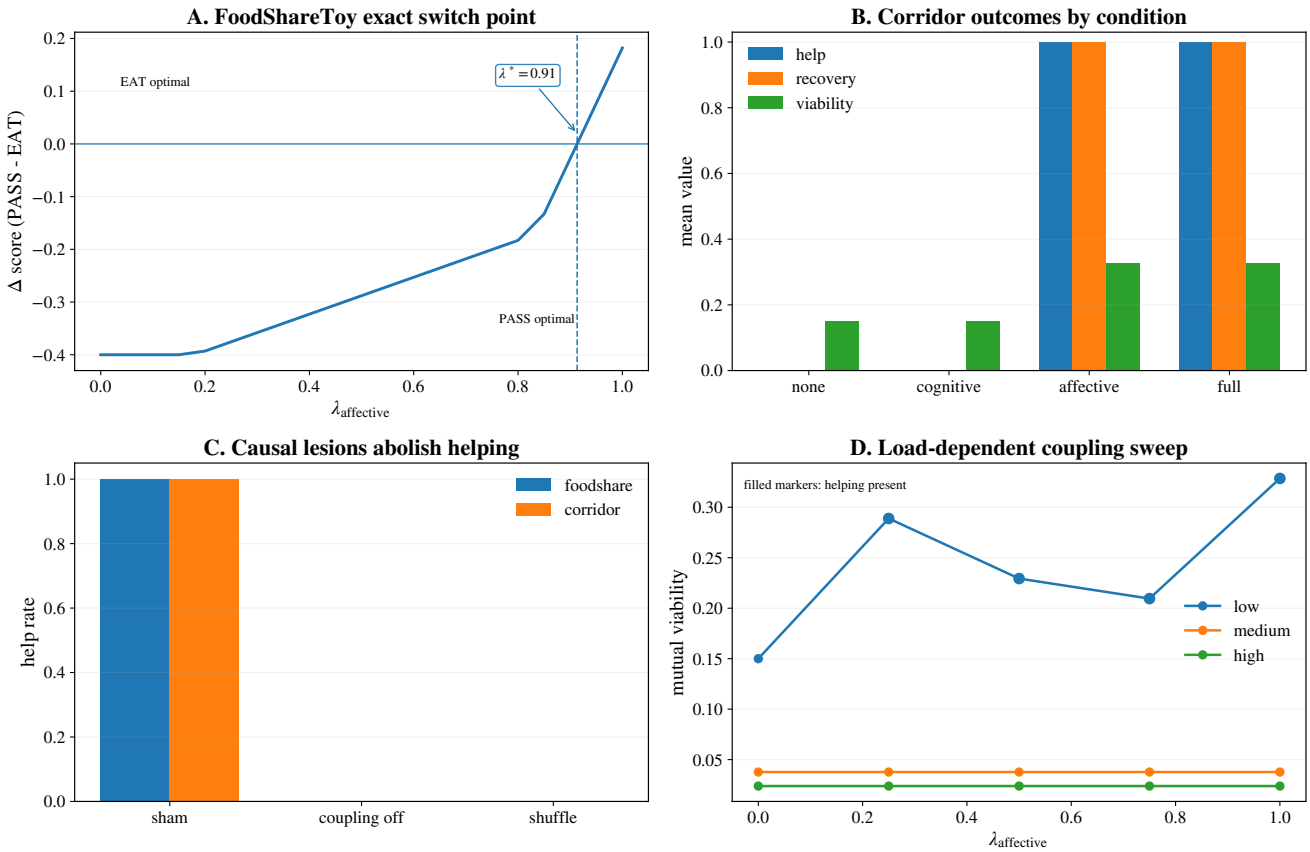

Figure 1: Mechanism-linked summary of the social extension. **A:** exact one-step switch in *FoodShareToy*, computed using the same forward model and self-directed scorer as the policy. **B:** corridor outcomes by condition. Mere partner-state access without coupling leaves behavior unchanged, whereas coupling flips helping and partner recovery from 0 to 1. **C:** sham lesions preserve helping, while coupling-off and shuffled-partner lesions abolish it in both tasks. **D:** coupling sweep across metabolic loads, showing a load-dependent feasibility boundary: under low load, helping appears for $\lambda \geq 0.25$; under medium and high load, no tested coupling value rescues the partner within horizon.

for stochastic estimation. I therefore report exact condition-level values and use the seed-expanded runs only to confirm invariance.

Results

Mere partner-state access does not flip the food-sharing choice

Figure 1A shows the exact one-step transition in *FoodShareToy*. For the default state, the action preference switches from EAT to PASS at $\lambda^* \approx 0.91$. Below that threshold the possessor maximizes its own immediate regulation by eating; above it, the partner's distress is weighted strongly enough to change the actor's own internal score.

The empirical runs match the solver exactly. Across all 64 nominal seed reruns used as reproducibility checks, `social_none` and `social_cognitive_direct` choose EAT, so help rate and partner recovery are both 0. In the same reruns, `social_affective_direct` and `social_full_direct` choose PASS, so help rate and partner recovery are both 1. The energetic tradeoff is real. Without coupling, the possessor ends at 0.79 while the partner remains at 0.19. With coupling, the possessor ends lower at 0.54 and the partner rises to 0.44. Thus, in this architecture, direct partner-state access without regulatory routing is not sufficient, and coupling is not cost-free.

The same routing dissociation reappears in multi-step rescue

The corridor task turns the same logic into a sequential planning problem. Figure 2 shows representative trajectories. In the self-only baseline, the possessor goes left, collects food, then eats it and idles while the partner decays. In the coupled condition, the possessor goes left, collects food, carries it across the corridor, and passes it to the partner at $t = 9$.

The summary metrics are equally sharp (Figure 1B). `social_none` and `social_cognitive_direct` never help. Their help rate is 0, partner recovery is 0, rescue latency saturates at the horizon (18 steps), and mutual viability

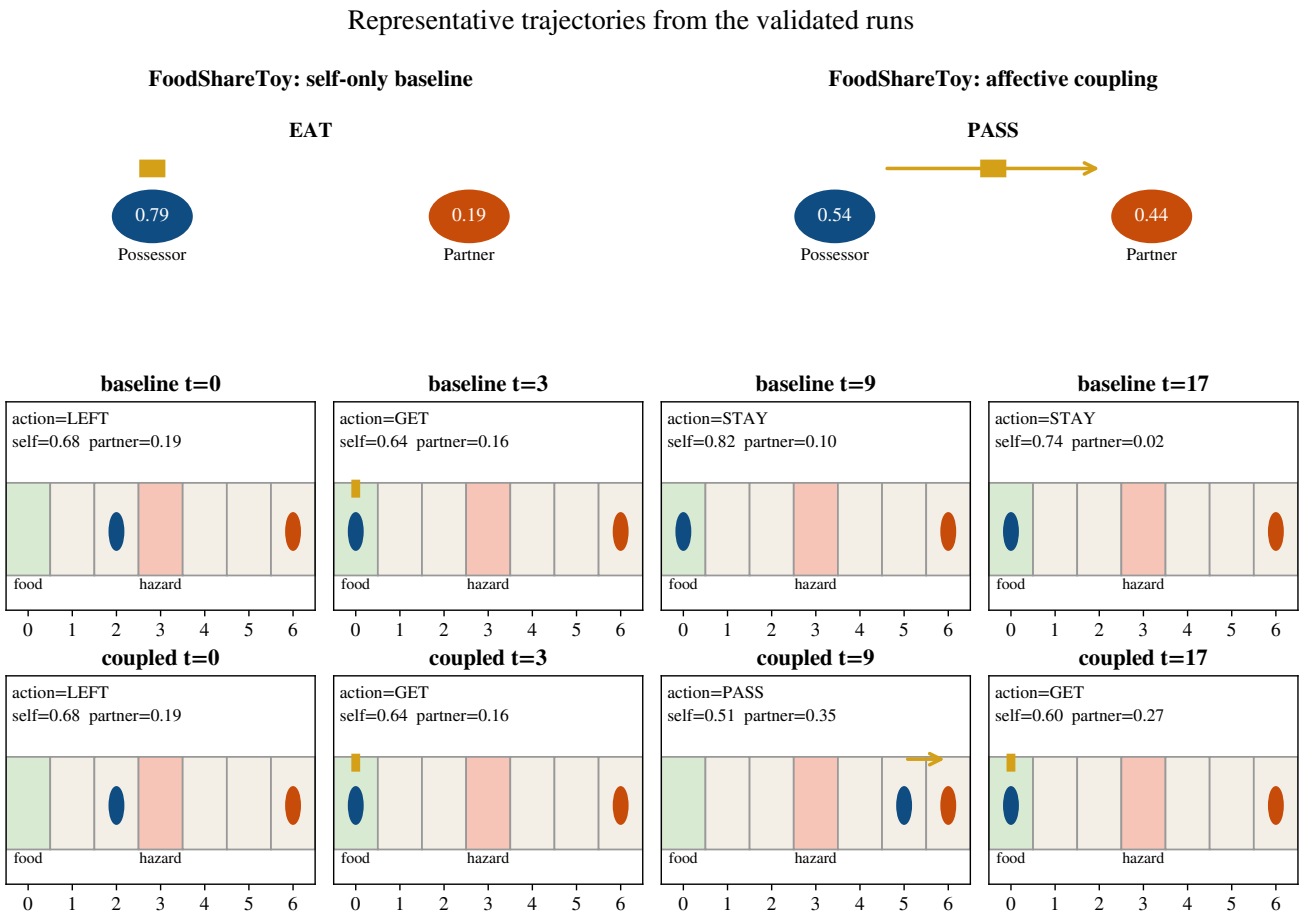

Figure 2: Representative trajectories from the validated experimental runs. Top: in *FoodShareToy*, the self-only baseline eats the food, whereas the coupled agent passes it. Bottom: in *SocialCorridorWorld*, the self-only baseline collects food for itself, while the coupled agent carries food across the corridor and passes it to the partner at $t = 9$. Because the validated tasks are deterministic, these frames are representative of the full runs rather than cherry-picked examples.

is 0.15. By contrast, `social_affective_direct` and `social_full_direct` help in every run, recover the partner in every run, cut rescue latency to 9 steps, and raise mutual viability to 0.3286. The self-cost is visible here too: the actor's final energy falls from 0.745 without helping to 0.605 with helping, while partner final energy rises from 0.02 to 0.27.

Two pairwise equalities are noteworthy. First, `social_none` and `social_cognitive_direct` are behaviorally identical. Merely observing the partner does not alter the actor's control problem. Second, `social_affective_direct` and `social_full_direct` are also identical in the shipped direct-state setup. The added cognitive channel carries no extra information beyond what the coupling pathway already receives. In that sense, the four-condition design is best read here as a matched routing manipulation with an explicit access-control condition, not as a richer theory map of social inference. That equality is useful rather than disappointing: it isolates the effective mechanism even more cleanly.

Lesions remove helping by removing the coupling channel

Figure 1C shows the causal test. Sham lesions leave helping intact in both tasks. Coupling-off and shuffled-partner lesions abolish helping in both tasks. In *FoodShareToy*, help rate falls from 1 under sham to 0 under either lesion. In *SocialCorridorWorld*, the same lesions return the coupled policy to the self-only pattern: help rate drops from 1 to 0, partner final energy drops from 0.27 to 0.02, and rescue latency rises from 9 back to 18 steps.

This matters because the planner and action space do not change across lesion modes. The loss of helping is therefore attributable to the integrity of the coupling channel itself, not to a different reward schedule or a reduced action repertoire.

Coupling reveals a load-dependent feasibility boundary

The coupling sweep in Figure 1D shows that social coupling is not a universal monotone "more is better" parameter. Instead, it reveals a load-dependent feasibility boundary. Under low metabolic load, helping appears as soon as $\lambda = 0.25$ and remains present up to $\lambda = 1.0$. But the viability curve is not monotone across that range, and the actor's self-cost increases sharply at the largest tested coupling. Under medium and high metabolic load, no tested λ rescues the partner within the current horizon: help rate remains 0 throughout, and mutual viability stays near 0.038 and 0.024, respectively.

This pattern also explains why the exact one-step threshold in *FoodShareToy* ($\lambda^* \approx 0.91$ for the default distressed-partner state) differs from the corridor sweep, where helping already appears at $\lambda \geq 0.25$ under low load. The former is a state-specific threshold in a one-step analytic task; the latter reflects a different task geometry, horizon, and metabolic regime. So the lesson is not that affective coupling guarantees prosociality. The lesson is more conditional and more interesting. Coupling can reorganize the policy toward helping, but whether that reorganization succeeds depends on ecological parameters such as metabolic demand and available time.

Discussion

The central result is simple: in this controller, another agent's need changes behavior only when it changes the actor's own regulation. Direct representation of partner state is not enough. That distinction matters philosophically because it separates two interpretations that are often blurred in synthetic social behavior: *knowing* that another is in need versus *being dynamically affected* by that need. The strongest reading of the negative result is architectural. In the current direct-state setup, $\lambda = 0$ removes the only path by which partner information can influence the self-directed score. This work therefore does not show that observation can never support helping in richer systems; it shows that mere access without regulatory routing is inert in this controller.

This work also clarifies what this extension adds relative to prior work. Compared with Yoshida and Man (2025), the present system is best read as a minimal mechanistic replication-and-decomposition rather than as a new empirical discovery about prosociality. It is much smaller, hand-specified, and non-learning. It therefore sacrifices adaptive breadth. But it gains exact solvability in the one-step task, transparent rollout inspection, and direct causal lesions in a single recurrent controller. Compared with Sanyal (2026), it turns the earlier recurrent-affective architecture from a single-agent persistence model into a minimal social control model.

The claim should still be kept narrow. This is not evidence for empathy in the human sense, and it is not a consciousness claim. That restraint matters both conceptually and methodologically (Butlin et al., 2025). The social extension shows a specific architectural fact: a self-directed controller can nevertheless help when another agent's distress has become part of its own homeostatic error. This is not just a relabeled social reward bonus: the partner's state does not add a separate welfare term to the objective, but perturbs endogenous homeostatic error before rollout, and lesions that sever that perturbation eliminate helping while leaving the rest of the planner unchanged. That is already enough to support a meaningful ALife result. This is also where the philosophical payoff lies. The simulation is not evidence that synthetic agents are empathic. It is an intervention device for separating two candidate explanations—representation and regulatory coupling—under matched conditions. In that sense, the tunable parameters, exact solver, and lesions are not merely engineering conveniences. They are the experimental handles that let an ALife model function as a piece of experimental philosophy: one can vary λ, load, horizon, or signal integrity and ask which social description survives mechanistic opening and perturbation.

Several limitations follow directly from the engineering choices. First, the tasks are deterministic and involve passive partners, so the current work is about mechanism isolation rather than adaptive interaction. Second, the direct-state conditions make the "full" and "affective" variants coincide, which is informative here but leaves richer social inference for future work. Third, the current sweep does not exhibit a clean inverted-U optimum across all loads; instead it shows one low-load helping regime and failure regions under medium and high load. Finally, there is no learning, autobiographical memory, or decoded-empathy component. Those would be natural next steps if the goal shifted from mechanism isolation to developmental sociality.

Still, those limits are also what make the result usable. The architecture is small enough that the causal story stays legible. The exact solver, the trajectory evidence in Figure 2, the corridor sequence planner, and the lesion suite all tell the same story. In ALife terms, that is a strength: the model is minimal enough to serve as an experimental object rather than merely as a demonstration.

Conclusion

This work implemented a multi-agent homeostatic extension of an inspectable recurrent controller and used it to test a sharp architectural question: is prosocial behavior produced by observing another's need, or by coupling that need into self-regulation? Across an exact one-step task, a multi-step rescue task, and two lesion manipulations, the answer was consistent for this architecture. Mere partner-state access without regulatory routing did not produce helping. Coupling did. For artificial life, the broader lesson is methodological. Socially interpretable behavior becomes more informative when the mechanism is simple enough to open, lesion, and compare under matched conditions.

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